

Concept Document: PAL-CleanX

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Track: Space Sustainability & Orbital Debris Remediation

1. Executive Summary

PAL-CleanX is an autonomous, AI-driven robotic platform designed to mitigate the threat of orbital debris in Low Earth Orbit (LEO). By integrating real-time computer vision (CNNs) and reinforcement learning (PPO), the system transitions debris removal from a high-cost manual task to a scalable automated service. The project combines advanced orbital research with a physical robotic arm prototype.

2. Problem Statement

LEO is congested with over 34,000 tracked debris fragments, threatening \$900B in global satellite infrastructure. Existing solutions are prohibitively expensive and rely on high-latency ground control, which fails to capture non-cooperative, tumbling objects.

3. Proposed Solution

A fully autonomous robotic system capable of "on-the-fly" decision-making. PAL-CleanX utilizes a hybrid capture payload—nets for small fragments and robotic arms for larger defunct satellites—eliminating the need for constant human intervention.

4. System Architecture & Technical Design

A. Components

- **Sensors:** LiDAR and high-resolution cameras for 360° awareness.
- **Processing Unit:** Onboard radiation-hardened CPU running CNNs and Kalman Filters.
- **Actuators:** Multi-jointed robotic arm (6-DOF) and Ion thrusters for maneuvering.

B. Data Flow

1. **Sensors:** Capture kinematic and spectral data.
2. **AI Model:** Classifies debris and predicts trajectory patterns.
3. **Control System:** Executes autonomous docking and capture.
4. **Disposal:** Guided de-orbiting for atmospheric incineration.

5. Technical Implementation

The project is validated through a dual-track approach:

- **Simulation:** High-fidelity orbital dynamics modeling in MATLAB/ANSYS.

- **Physical Prototype:** A functional 6-DOF robotic capturing arm demonstrating the mechanical feasibility of the gripper system.

6. SDG Alignment

SDG 9: Industry, Innovation, and Infrastructure

Space is critical global infrastructure. PAL-CleanX protects the orbital environment, ensuring the long-term sustainability of the space industry and its terrestrial benefits (GPS, Telecom).

7. Scalability & Future Work

- **Swarm Robotics:** Coordination of multiple units for large-scale cleanup.
- **Future Work:** Microgravity testing and refining docking accuracy to >99%.

8. Conclusion

PAL-CleanX offers a pragmatic, AI-powered solution to the space debris crisis, securing humanity's continued access to space through proactive and sustainable innovation.